

Name: _____ Mark: _____ /72

G10 Science - Class 2 Homework

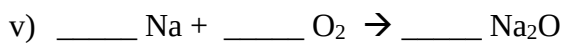
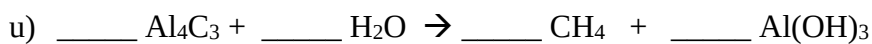
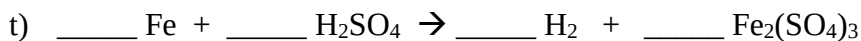
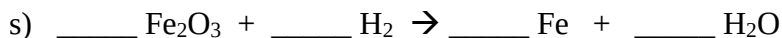
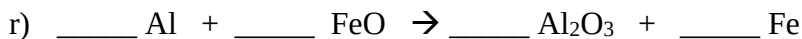
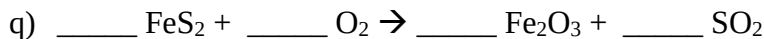
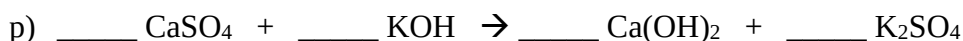
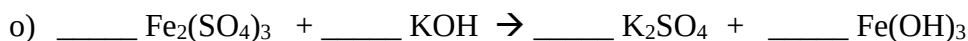
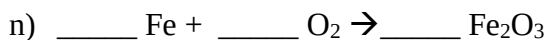
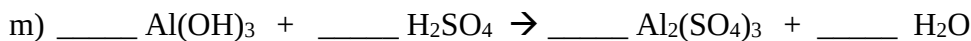
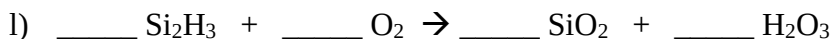
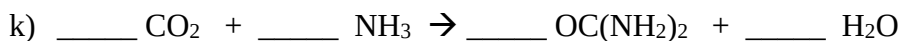
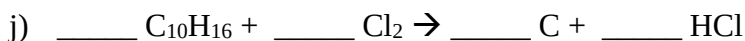
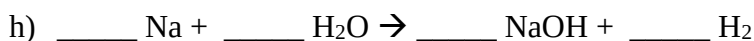
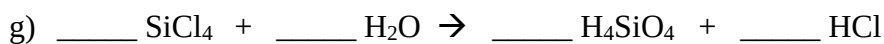
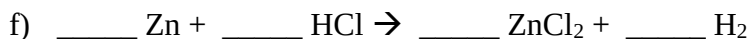
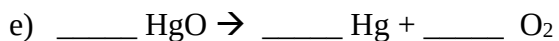
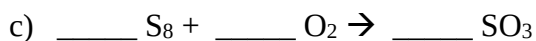
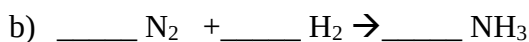
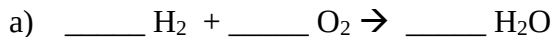
1. Consider the following reaction: $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$

- a) Label the reactants and products in this reaction. **[2 marks]**
- b) Write the chemical name for the compounds that are dissolved in water. **[3 marks]**
- c) Write the chemical name for the precipitate. **[1 mark]**

2. Write the balanced chemical equation for the following: **[14 marks]**

- a) Aluminum + Copper(II) chloride $\rightarrow$ Aluminum chloride + Copper
- b) Zinc + Sulfur $\rightarrow$ Zinc sulfide
- c) Silicon dioxide + Hydrogen fluoride $\rightarrow$ Silicon tetrafluoride + Water
- d) Phosphorus + Oxygen + Water $\rightarrow$ Hydrogen phosphate
- e) Ammonia + Nitrogen monoxide $\rightarrow$ Nitrogen + Water
- f) Silver nitrate + Potassium chromate $\rightarrow$ Silver chromate + Potassium nitrate

3. Balance the following chemical equations: [22 marks]



4. Classify the following chemical equations as synthesis (syn), decomposition (dec), single displacement (SD), double displacement (DD), or combustion (comb). **[5 marks]**

_____ a) Ammonia + Sulfuric acid (H_2SO_4) $\rightarrow$ Ammonium sulfate

_____ b) Iron + Copper(II) sulfate $\rightarrow$ Iron(III) sulfate + Copper

_____ c) Potassium carbonate + Barium chloride $\rightarrow$ Potassium chloride + Barium carbonate

_____ d) Aluminum sulfate $\rightarrow$ Aluminum oxide + Sulfur trioxide

_____ e) Ethane (C_2H_6) + Oxygen $\rightarrow$ Carbon dioxide + Water

5. Write a balanced chemical equation for the each of the reactions in Question (4). **[10 marks]**

a) _____

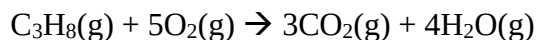
b) _____

c) _____

d) _____

e) _____

6. The following chemical equation represents the combustion of propane in a sealed container:



Use the data below to determine the mass of oxygen consumed in the reaction. Show all work.
[3 marks]

Item	Mass (g)
Container only	50.0
Propane and container	61.8
Carbon dioxide, water, and container	104.8

7. Predict the products of the following complete combustion reactions and balance: **[5 marks]**

a) Pentane (C_5H_{12})

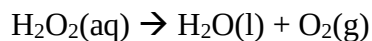
b) Diethyl ether ($\text{C}_4\text{H}_{10}\text{O}$)

c) Urea ($\text{CH}_4\text{N}_2\text{O}$)

d) Benzene (C_6H_6)

e) Propanol ($\text{C}_3\text{H}_7\text{OH}$)

8. Hydrogen peroxide is a clear, colourless liquid used to disinfect cuts. The chemical equation is as follows:



a) Balance and classify this reaction. **[2 marks]**

b) Explain the difference between the state symbols (aq) and (l). **[2 marks]**

c) Predict what you would observe in the test tube as the reaction proceeds. **[2 marks]**

d) How would you know when your reaction is complete? **[1 mark]**